

User Study – ML2++ (Textual DSL)

Instructions: This study is for non-coding users. You will **not** write code or model instances. Use a pre-built ML2++ model and click through the workflow: Save Model → Generate Code → Run Code → Show Forecasting Plots.

Participant ID: P7 Use case(s): ☐ U1 River Flow ☐ U2 Smart-Home Energy
☒ U3 Solar Energy

Section A – Background (answer once)

This section helps us understand your experience level so we can analyze results fairly.

A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☒ Master's ☐ Doctorate ☐ Other: _____

A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☐ MDE Practitioner ☒ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: _____

A3. Self-rated expertise (*tick one per row*)

	Beginner	Intermediate	Advanced	Expert
IoT/CPS development	☐	☒	☐	☐
Model-Driven Engineering (MDE)	☐	☐	☒	☐
Machine Learning / Data Science	☐	☐	☒	☐

A4. Prior experience (*used DSLs/modeling tools/ML frameworks before?*) ☒ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use case*)

This B-section is for: ☐ U1 River Flow ☐ U2 Smart-Home Energy ☒ U3 Solar Energy

If you did more than one use case, copy this whole Section B for each one and fill it separately.

B1. Task Completion & Ease of Use

B1. Were you able to finish the use case?

☐ None ☐ Partial ☒ Complete

B2. Ease by workflow step (*1=Very easy → 5=Very difficult*)

Step	1	2	3	4	5
Whole workflow (end-to-end)	☐	☐	☒	☐	☐
Data preprocessing (preview/EDA)	☐	☒	☐	☐	☐
Model configuration (preset options)	☐	☐	☐	☒	☐
Run/Execution (logs understanding)	☐	☒	☐	☐	☐
Model evaluation (metrics/plots)	☐	☐	☒	☐	☐

B3. Ease of ML2++ elements (*1=Very easy → 5=Very difficult*)

Element	1	2	3	4	5
Data types & objects	☐	☒	☐	☐	☐
Things & behaviours (statecharts)	☐	☐	☐	☒	☐
Ports & properties	☐	☐	☐	☒	☐
Preset model configuration options	☐	☐	☒	☐	☐
Configuration (instances/connectors)	☐	☐	☒	☐	☐

B2. Learning & Usability**B4. Learning curve** (*1=Very easy → 5=Very difficult*)☐1 ☐2 ☒3 ☐4 ☐5**B5. Intuitiveness** (*1=Very intuitive → 5=Not intuitive*)☐1 ☐2 ☐3 ☒4 ☐5**B6. Hardest part to learn (briefly)**Ports and properties

B3. Debugging & Error Handling**B7. Error frequency** (*0=None, 1=Few, 2=Moderate, 3=Many*)☐0 ☐1 ☒2 ☐3**B8. Clarity of error messages** (*1=Not clear → 5=Very clear*)☐1 ☐2 ☐3 ☒4 ☐5**B9. Debugging effort** (*1=Easy → 5=Hard*)☐1 ☐2 ☒3 ☐4 ☐5**B4. Code Quality, Maintainability, Flexibility***Refers to the generated code you viewed (no editing required).***B10. Generated code quality** (*1=Poor → 5=Excellent*)☐1 ☐2 ☐3 ☒4 ☐5**B11. Maintainability** (*1=Very difficult → 5=Very easy*)☐1 ☐2 ☐3 ☐4 ☒5**B12. Flexibility / Customisation** (*1=None → 5=Very flexible*)☐1 ☐2 ☐3 ☒4 ☐5

Section C – Overall Assessment (ML2++ only)**C1. Overall satisfaction** (*1=Very dissatisfied → 5=Very satisfied*)☐1 ☐2 ☐3 ☒4 ☐5**C2. Would you use ML2++ for your next similar project?**☒ Yes ☐ No ☐ Undecided *Why?* Quickness in changing new requirements

C3. Feature effectiveness (*1=Ineffective → 5=Very effective*)

Feature	1	2	3	4	5
Time-series forecasting models	☐	☐	☐	☒	☐
Automatic preprocessing & plots	☐	☐	☐	☒	☐
Training configuration options	☐	☐	☒	☐	☐
Multi-day forecasting accuracy	☐	☐	☐	☒	☐

C4. Favourite feature: Time series forecasting

C5. Least favourite feature: Training configuration

C6. Missing features you would like: _____

Section D – Open Feedback

D1. Barriers to adoption (*what would stop you from using ML2++ in production?*)

Need to process unstructured datasets (e.g. new)

D2. One improvement you would prioritize

Supporting pipeline definition to bigdata processing

Scale anchors used throughout for consistency:

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.